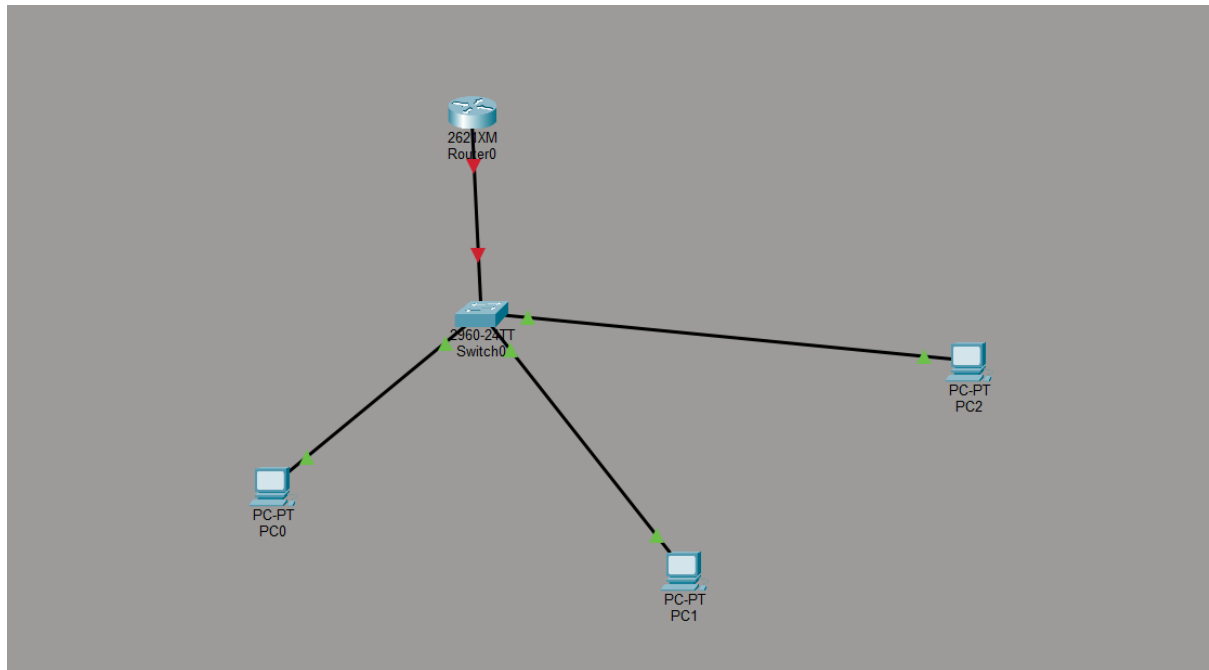


Network devices

- 1) Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.



- 2) Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

Network Device	Primary Function	Real-World Use Case	Example App/Service
Router	Connects different networks (e.g., home network to the Internet) and directs data between them.	Provides Internet access to all devices at home so you can browse, stream, and communicate online.	YouTube, Netflix, Google Chrome
Switch	Connects multiple devices within the same Local Area Network (LAN) and forwards data to the correct device.	Connects desktop PCs and gaming consoles for fast LAN gaming or file sharing in a home or school network.	Steam LAN gaming, Windows File Sharing
Hub	Connects multiple devices in a LAN but broadcasts data to every connected device.	Used in small test labs or educational demonstrations to show how network traffic is shared. (Rarely used today due to lower efficiency.)	Classroom networking practice
Wireless Access Point (AP)	Provides wireless (Wi-Fi) access to a wired network.	Allows smartphones, tablets, and laptops to connect to Wi-Fi for streaming, online classes, and social media.	YouTube, Zoom, Spotify, WhatsApp

- 3) Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).
- I open the Zomato app on my phone and select a restaurant to order food. My phone first connects wirelessly to the Access Point (AP), which provides the Wi-Fi connection. The access point sends my request to the Switch, which forwards the data to the correct device on the local network. The data then reaches the Router, which directs it from my home network to the Internet. The router sends the request to Zomato's servers, where my order is processed. The confirmation travels back through the Internet to the Router, then to the Switch, then to the Access Point, and finally reaches my phone, where I see that my food order has been successfully placed.

Roles of the network devices:

- **Access Point (AP):** Connects my phone to the Wi-Fi network.
- **Switch:** Transfers data efficiently between devices on the local network.
- **Router:** Connects my home network to the Internet and sends/receives data from Zomato's servers.

- 4) Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.

Hint: Think about devices like switches, powerline adapters, or WiFi extenders.

Network Device	How It Works in This Setup	Advantage
Powerline Adapter	Plug one powerline adapter into a wall socket near the WiFi router and connect it to the router with an Ethernet cable. Plug the second adapter into a wall socket near the gaming PC and connect it to the PC with an Ethernet cable. The adapters use your home's electrical wiring to carry the network signal.	Provides a stable wired connection without running a long Ethernet cable through the house.
WiFi Extender (with Ethernet Port)	Place the WiFi extender between the router and the gaming PC. The extender connects wirelessly to the router and then provides an Ethernet port that the gaming PC can connect to using an Ethernet cable.	Extends WiFi coverage and allows an Ethernet-only device to connect to the wireless network.